

End-to-end Quality of Experience Evaluation for HTTP Adaptive Streaming

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ABSTRACT

Exponential growth in multimedia streaming traffic over the Internet motivates the research and further investigation of the user's perceived quality of such services. Enhancement of experienced quality by the users becomes more substantial when service providers compete on establishing superiority by gaining more subscribers or customers. Quality of Experience (QoE) enhancement would not be possible without an authentic and accurate assessment of the streaming sessions. HTTP Adaptive Streaming (HAS) is today's prevailing technique to deliver the highest possible audio and video content quality to the users. An end-to-end evaluation of QoE in HAS covers the precise measurement of the metrics that affect the perceived quality, e.g., startup delay, stall events, and delivered media quality. Mentioned metrics improvements could limit the service's scalability, which is an important factor in real-world scenarios. In this study, we will investigate the stated metrics, best practices and evaluations methods, and available techniques with an aim to (i) design and develop practical and scalable measurement tools and prototypes, (ii) provide a better understanding of current technologies and techniques (e.g., Adaptive Bitrate algorithms), (iii) conduct in-depth research on the significant metrics in a way that improvements of QoE with scalability in mind would be feasible, and finally (iv) provide a comprehensive QoE model which outperforms state-of-the-art models.

CCS CONCEPTS

- **Information systems** → **Information systems applications**;
- **Multimedia information systems** → *Multimedia streaming*.

KEYWORDS

HTTP Adaptive Streaming; Quality of Experience; Subjective Evaluation; Objective Evaluation; Adaptive Bitrate; QoE model.

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1 INTRODUCTION

Over 66% of mobile data traffic in 2020 was video content. A share that is predicted to increase to 77% by 2026 [6]. Moreover, the COVID-19 pandemic affected the number of viewers streaming video and viewing hours even more in 2020 [2]. A report published by Cisco Systems Inc. [4] predicts that video streaming services would utilize 82% of the consumer internet traffic in 2022. Highlighted by technical reports and predictions stated above, we should expect further video content traffic growth soon.

Among all the video streaming techniques available today, HAS, with the two most commonly implemented media delivery formats, HTTP Live Streaming (HLS) [1] and Dynamic Adaptive Streaming over HTTP (MPEG-DASH) [7], is predominant. The primary added value that HAS technique brings is adapting the streaming session to the media delivery link constraints. This advantage of HAS is made possible by breaking the media file into multiple audio and video segments or chunks of the same duration, often between 2 to 10 seconds. Instead of delivering a single media file to the client, a manifest file that acts as a playlist and defines the media streams' available adaptation sets and representations will be sent to the media player. Leveraging this video streaming technique, media players would have the option to select the best possible quality for the next segment to have a smooth media streaming session. This decision is made based on the real-time network conditions and property observation done by the client or media player. The logic behind making such decisions in media players called the Adaptive BitRate (ABR) Algorithm, considers various available resources on end-user devices. An ABR algorithm's primary function is to provide the highest possible QoE for the end-user considering the available resources, e.g., buffer size, available bandwidth, network latency, or a combination thereof.

Since the media players and their ABRs formulate factors to select the proper media quality in real-time and internet link properties are volatile, the measurement of end-user perceived QoE is not a straightforward task. One should carefully observe the variables such as startup delay, the number, and duration of stall events in a streaming session alongside the provided media quality to calculate the perceived QoE accurately. Quality models like ITU-T P.1203 [10] algorithmically predict a Mean Opinion Score (MOS) by considering those variables and the actual media quality factors.

HAS can be divided into *Content Provisioning*, *Content Delivery*, *Content Consumption*, and *End-to-end Aspects*. The *End-to-end Aspects* cover the other three categories from a higher point of view, which means less granular detail of other categories will be

included, with integration and complete system functionality, being the main consideration.

This study will cover the *End-to-end Aspects* of HAS and investigates the QoE from the whole streaming session point of view. A holistic approach to HAS with QoE being the primary consideration. The rest of this paper is organized as follows. The current state of the art in this field will be reviewed in the Section 2. In Section 3, we will discuss the research questions and describe the proposed methodology in Section 4 and the publications will be listed in Section 5. Finally, in Section 6, we conclude the paper with a summary of the presented study.

2 RELATED WORKS

Brunnström *et al.* [3] provided a broad and in-depth study of QoE definition, its application areas, the factors influencing QoE, and the features of the defined QoE. This thesis further investigates the defined QoE by mentioned work concerning the End-to-end aspect of the HAS specifically.

Seufert *et al.* [11] did a comprehensive survey of QoE-related works in human-computer interaction and networking domains, structured according to the QoE impact of video adaptation. They have revisited the subjective studies that cover QoE aspects of adaptation dimensions and strategies. An overview of the current state of the art has been provided, giving insight into perceptual QoE influence factors.

In the context of HAS, QoE has been further refined and surveyed by Seufert *et al.* ; [11] concerning a taxonomy of HAS QoE influence factors clustered into (i) perceptual and (ii) technical influence factors. Moreover, authors have addressed QoE management in the context of ongoing developments such as moving toward softwarized networks, exploiting big data analytics and machine learning, and the steady rise of new and immersive services (*e.g.*, augmented and virtual reality). They provided an understanding of the implications of such paradigm shifts in terms of new approaches in QoE modeling and the need for novel QoE monitoring and management infrastructures.

In general, more recent surveys addresses concepts and challenges for QoE management of multimedia services [9, 12] by categorizing them into (i) modeling, (ii) monitoring, and (iii) management.

3 RESEARCH QUESTIONS

This section will introduce the three research questions that we will investigate accordingly. Following are the research questions with a brief description for each.

- (1) How to define, implement, and deploy end-to-end QoE for HAS deployments at scale?

We will investigate the possibilities of how to assess a video streaming session to understand its characteristics and implement improvements on the end-to-end aspects toward a better perceived QoE. Scalability plays an important role in perceived QoE in video streaming, which means the number of concurrent viewers or clients could significantly impact each session's QoE. Therefore, we will develop our novel solutions and tools so that the system could scale up the

clients' quantity in real-time and mimic experiments with the same dimensions as in a real-life setting.

- (2) How to develop a QoE-driven, context-aware, and collaborative HAS environment?

As a hypothesis, we believe that the results of the QoE assessment of video streaming sessions can give useful insights into the perceived QoE by actual end-users for each category of video content, *e.g.*, animation, action, game events. Keeping this hypothesis in mind, we will investigate how to improve the perceived QoE for each content type.

- (3) How to utilize various information from end-to-end aspects to enhance HAS deployments?

For instance, the *HAS's content delivery* aspect refers to the network links between the network nodes *e.g.*, the connection between the client machine and the content server. The properties of such links can differ in different conditions and strongly relate to the user's level of annoyance or delightfulness. We will investigate how these properties from end-to-end aspects affect the video streaming session QoE and propose a solution to improve the QoE in HAS.

4 RESEARCH METHODOLOGY

We use an Iterative and Incremental Development (IID) approach [8] to (i) design, (ii) develop, and (iii) evaluate proposed solutions, by which means the progress toward the defined goals would be added incrementally in each round of iterations over the mentioned phases and also evaluated iteratively. A system will be developed in IID methodology through repeated cycles (iterative) and in smaller portions (incremental). This approach enables us to learn during the development of the earlier parts or versions of the solution and apply modifications to the next steps accordingly if needed. This is also associated with iterations an engineer expects over (i) state requirements, (ii) state specifications, (iii) design and implement the system, (iv) and test the system [5].

We will use an experimental approach, where all novel tools and methods will be evaluated and validated using controlled experiments through objective or subjective evaluations. Published papers that are listed in the Section 5 are examples of this experimental approach. The mentioned approach will be taken into consideration toward the end of this thesis. With empirical research methods, hypotheses will be formed based on the research questions from Section 3. These hypotheses will be measured quantitatively, rigorously analyzed, and evaluated according to established research methods. Furthermore, finally, the outcomes and results of this thesis will be reported by presenting quantitative research results.

5 PUBLICATIONS AND ON-GOING AND FUTURE WORK

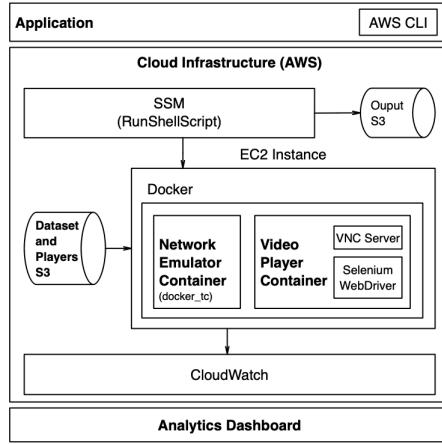
Table 1 provides a list of papers that the first author is the author of this paper and are designed to investigate further the research questions stated in Section 3. Listed publications include the studies that will be worked on in the near future. Also, in Table,1 possible targets for future works are listed. Here a brief overview of each paper is provided as follows.

Cloud-based Adaptive Video Streaming Evaluation (CAdViSE) [14] is a framework for the automated testing of adaptive media

Table 1: List of Related Publications and Possible Target Conferences and Journals

Row	Title	Venue	Rank ¹	Status	Ref.
1	CAdViSE: Cloud-based Adaptive Video Streaming Evaluation framework for the automated testing of media players	MMSys'20: Proceedings of the 11th ACM Multimedia Systems Conference	B3 Qualis 2012	Published	[14]
2	Understanding Quality-of-Experience of Heuristic-based HTTP Adaptive Bitrate Algorithms	NOSSDAV'21: The 31st edition of the Workshop on Network and Operating System Support for Digital Audio and Video	A2 Qualis 2012	Published	[13]
3	INTENSE: In-depth Studies on Stall Events and Quality Switches and Their Impact on the Quality of Experience in HTTP Adaptive Streaming	A high quality journal	-	Under Review	-
4	A journal extension of the publication number 2	ACM Transactions on Multimedia Computing, Communications, and Applications	Q1 2021	Targeted (in progress)	-

¹ <http://www.conferencerranks.com/>, accessed Jun. 15, 2021.

**Figure 1: CAdViSE System Architecture [14].**

players. The paper aims to demonstrate a test framework that can be instantiated in cloud infrastructure, examines multiple media players and their ABR performance in different network attributes at defined points of the experiment time, and finally concludes the evaluation with visualized statistics and insights into the results.

Figure 1 depicts the components and architecture of the CAdViSE. Using Cloud infrastructure, the testbed is maintaining scalability, and also the source code is made available in a GitHub repository¹ which makes the reproducibility feasible. The application component consists of *bash* scripts that get executed in different phases of running the evaluation: (i) a runner script, (ii) an initializer script, and (iii) a starter script. A *python* script that controls the Selenium WebDriver² is also included in this component. The cloud infrastructure is considered another component of the system. The AWS Elastic Compute Cloud³ (EC2) instances are managed and controlled by the AWS Systems Manager⁴ (SSM), and the log files (output or errors) of the command executions will be stored in an

Amazon Simple Storage Service⁵ (S3) bucket. Useful information such as CPU load or network usage of the EC2 instances is gathered in Amazon CloudWatch⁶ and can be monitored in real-time on the AWS Management Console.

Using the previous paper's introduced framework, we have conducted objective and extensive subjective evaluations with crowd-sourced participants and investigated the Mean Opinion Score (MOS) of the streaming sessions. The second paper's [13] main contribution is subjective evaluation analogy with objective evaluation for well-known heuristic-based Adaptive BitRate (ABR) algorithms. We have studied how well several known ABR algorithms make bitrate decisions with different network profiles, ramp-up, ramp-down, fluctuating, and stable. We have examined the correlations and contradictions between the ABR algorithms' subjective and objective evaluations. Figure 2 shows the measured values of the important metrics, stall events duration, and startup delay for the studied ABR algorithms.

Based on the findings from previous publications, we have conducted another experimental research to investigate further the significant QoE metrics (e.g., Stall events, Startup delay, Media quality). The third paper, introduced in Table 1, studies the open issues related to the QoE for HAS, notably (i) the minimum noticeable duration for stall events in HAS as well as (ii) the correlation between the media quality and the impact of stall events on QoE; (iii) the end-user preference regarding multiple shorter stall events versus a single longer stall event; and (iv) the end-user preference of media quality switches over stall events. The paper presents in-depth studies on the impact of stall events and quality switches on QoE from objective and subjective evaluation perspectives. It presents the correlation between the two types of evaluations. The findings documented in that paper can be used as a baseline for improving ABR algorithms and policies in HAS.

6 SUMMARY

In this paper, we have presented our research questions designed toward end-to-end evaluation of QoE in HAS. We have briefly reviewed the related works and introduced the research methodology

¹<https://github.com/cd-athena/CAdViSE>, accessed Jun. 14, 2021.

²<https://www.selenium.dev/documentation/en/webdriver/>, accessed Feb. 26, 2020.

³<https://docs.aws.amazon.com/ec2/>, accessed Feb. 26, 2020.

⁴<https://docs.aws.amazon.com/systems-manager/>, accessed Feb. 26, 2020.

⁵<https://docs.aws.amazon.com/s3/>, accessed Feb. 27, 2020.

⁶<https://docs.aws.amazon.com/cloudwatch/>, accessed Feb. 26, 2020.

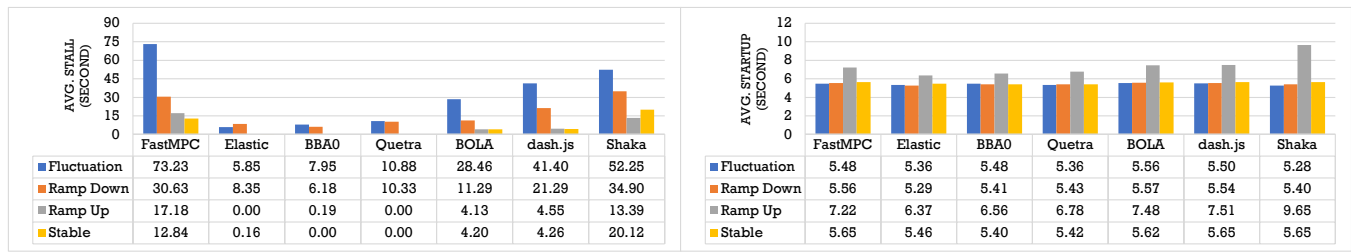


Figure 2: Avg. Stall Duration (left) and Avg. Video Startup Delay (right) of the ABR Algorithms Examined under Different Network Conditions [13].

used toward the end of this study. Our publications and ongoing works have been discussed, and we have stated the future work planned to be worked on.

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